

What Would You Add?

Four numbers, five complaints, and three proposals

Duration: 45 to 60 minutes

Group size: 3

Output: One Decision Sheet per group plus a 75 second report out

Module context: Fine-Tuning and Retrieval-Augmented Generation. You now have the full toolkit: stage-specific metrics from Segment 4, and the complexity patterns from Segment 5. This is where you use them to make an architecture decision under a deadline.

1. How this session runs

You will work in a group of three. Three is deliberate. Nobody gets to be a passenger, every position must be held by someone, and a group of three cannot hide a disagreement behind a majority. In your roles, you will walk through and respond to each round (as described below).

THERE ARE HINTS AND USING THEM IS NOT CHEATING Section 15 has a stuck ladder for the five hardest questions, three levels each. Take one level at a time and stop as soon as it unsticks you. Reading all three levels for a question you were about to solve wastes the part of this that teaches you something.

2. Your role

Each person takes one role. Roles are not decoration. Each one owns a different definition of the word better, and the friction between those definitions is the discussion.

Role	You own	The question you keep asking
Retrieval engineer	The index, the chunking, and the recall number	Is the right chunk reaching the model, and if not, where does it get lost?
Serving and latency owner	p95, the pager, and the fact that every pattern is a tax on every query	What does this cost on every single request, forever?
Evaluation lead	The eval set, the judge, and whether any number in this room means anything	How do we know, and how would we know we were wrong?

3. Where you are

It is four weeks after the four tickets. Those are closed. The citation check ships on every response, the Slack service loads the adapter now, and the chunk that split the zoning panel heading from its body has been repaired.

More importantly, you built an evaluation harness. For the first time, Cordwell Assist has numbers. These are the first ones it produced.

Metric	Value	Stage
Context recall	0.61	Retrieval
Context precision	0.84	Retrieval
Faithfulness	0.93	Generation
Answer relevancy	0.88	Generation

Users are still complaining. Your director has asked what you are going to add and has heard the word reranker from a vendor. The change freeze starts in three weeks.

YOUR ASSIGNMENT Decide what you would add, what you would measure to confirm it worked, and what you would refuse to add. Every one of those three must be defended with a number from this packet.

4. Reading the four metrics

Reference, not answer. The last column is the one people skip and it is the one that matters most here.

Metric	What it measures	A low value means	A high value does NOT mean
Context recall	Did the retrieved chunks contain the information needed to answer	The index or the query is failing you, before the model ever sees anything	That the answer will be good. Recall is silent about what the model then does with what it received.
Context precision	Were the relevant chunks ranked near the top	Ranking is noisy and the good chunk is buried	That a relevant chunk was in the set at all. Precision is silent about what was never retrieved.
Faithfulness	Is every claim in the answer supported by the retrieved context	The model fabricated something	That the answer was useful. A model handed poor context that faithfully replies "I do not have that" scores near perfect.
Answer relevancy	Does the answer address the question that was asked	The answer drifted off topic	That the answer was correct, or grounded, or complete.

5. The system as it runs today

Item	Value
Shape	Naive RAG. Embed the query, search, stuff the top chunks, generate. No reranker, no query transformation, no second pass.
Retriever	Top k of 4.

Item	Value
Corpus	About 18,400 support documents, chunked at 1024 tokens with no overlap.
Volume	About 9,600 assisted sessions in the last 30 days.
Generation	Small instruct model with the Cordwell adapter, running locally. 256 max new tokens.
Latency, p50	2.9 seconds end to end.
Latency, p95	6.2 seconds end to end.
Internal latency target	p95 under 4 seconds. Currently missed.
Team	Two engineers and a half-time SRE.
Calendar	The change freeze starts in three weeks and runs through January.

Where the 2.9 seconds goes

embed the query	~0.015 s		
vector search	~0.040 s		
assemble the prompt	~0.005 s		
generate the answer	2.80 s	(p50)	6.10 s (p95)

end to end	2.86 s	(p50)	6.20 s (p95)

Retrieval is roughly two percent of the budget. Generation is the rest. Any pattern that adds a model call is expensive. Any pattern that adds a search is close to free. Hold that shape in your head for Round 2.

6. How those four numbers were produced

Read this section slowly. Round 1 is built on it.

Item	Detail
Eval set size	40 queries.
Who wrote them	One engineer, in one afternoon, by reading through the corpus and writing a question for each document she opened.
Ground truth	For each query she marked the one chunk she considered correct. One gold chunk per query.
Unanswerable split	None. Every query in the set has an answer somewhere in the corpus.
Judge	The same small local model that generates the answers.
Value of k	All four numbers were measured at k of 4. No metric has ever been measured at any other k.
When it ran	Once, three weeks ago, on a Tuesday afternoon.

Item	Detail
Where it lives	A notebook on that engineer's laptop. Not in version control.

7. What users are complaining about

92 complaints across about 9,600 sessions in the last 30 days, clustered by a support lead who read all of them.

Count	Cluster	Representative wording
31	Had to rephrase before it found anything	"Took me three tries to word it right."
24	Does not handle questions with two parts	"Asked about breaker size and mounting height. Got one of them."
19	Too slow	"Faster to just open the manual myself."
11	The answer cuts off	"It stops mid-sentence."
7	Says it cannot find things I know are in there	"That document definitely exists."

8. The patterns, and what each one moves

Reference card. This is scaffolding, not an answer. Knowing what a pattern does is the easy half. Knowing whether this system needs it is the discussion.

Pattern	What it does	Moves	Does NOT move	Added latency per query
Query rewriting	Rephrases the query into the vocabulary the index actually uses	Context recall	Precision, faithfulness	~0.3 s with a model call, near zero with rules
Multi-query expansion	Generates several query variants, searches each, merges the results	Context recall	Faithfulness	~0.5 s
Decomposition	Splits a complex query into sub-questions and retrieves for each	Context recall on multi-part queries	Single fact recall	~2.8 to 5.6 s
HyDE	Generates a hypothetical answer, embeds that, searches with it	Context recall on vocabulary mismatch	Coverage gaps	~1.3 s
Reranking	Retrieves a wide candidate set cheaply, then reorders it with a cross-encoder	Context precision, and what reaches the model at a fixed k	What is in the candidate set to begin with	~0.3 s for 50 candidates
Self-reflective generation	A second pass checks whether the answer is supported, then revises	Faithfulness	Recall, precision, latency in any good direction	~2.8 to 5.6 s

Pattern	What it does	Moves	Does NOT move	Added latency per query
Agentic retrieval	The model decides how much to retrieve, with which filters, and whether to search again	Potentially all of them	Debuggability, in the wrong direction	Unbounded

9. Round 0: your first call

Silent. Individual. Do not discuss.

Write four things, then stop.

- The metric you think is the outlier.
- The one pattern you would add.
- Your confidence, one to five.
- The one thing you would want to know before committing to that.

10. Round 1: do you believe the numbers

Group. No architecture yet.

The rule this round: nobody proposes a pattern. If someone starts, the skeptic stops them. You are auditing the evidence you would be building on.

- 1.1** One of the four numbers is the outlier. Say which. Then answer the harder half: what do you learn from the combination that you could not learn from any one of them alone? Specifically, what does a faithfulness of 0.93 sitting next to a context recall of 0.61 tell you about what the model is doing?
- 1.2** Go through Section 6 line by line. For each line, decide whether it could push a number higher than reality, lower than reality, or make it meaningless. At least one line inflates a number and at least one line makes a different number close to uninterpretable. Name both and say which direction each one runs.
- 1.3** Map each of the five complaint clusters in Section 7 onto one of the four metrics. Two of the five do not map onto any of them. Which two, roughly how many users do they represent, and what does that tell you about a plan built only from the metric table?
- 1.4** 0.61 is an average over 40 queries. Name the two or three dimensions you would break it down by and predict what you would see. Write the prediction down. You will be held to it later.

Section 7 has already done part of this segmentation for you, using a completely different data source. That is a hint, not an answer.

- 1.5** Which is the cheaper next step: adding a pattern, or improving the measurement? Argue whichever side your group did not instinctively pick, for two minutes, before you decide.

11. Round 2: the bake-off

Group. Proposals are assigned, not chosen.

Three proposals are on the table. Each of you argues one, and it is deliberately not the one your role would reach for. There is a fourth option that nobody argues for, on purpose.

If your role is	You argue for	Which means you will have to
Serving and latency owner	Query transformation	Make the case for a pattern that adds work to every request
Evaluation lead	Reranking	Make the case for buying something before the measurement is solid
Retrieval engineer	Self-reflective generation	Make the case for a fix that lives entirely on the generation side
Nobody	Option four: add no pattern this quarter. Spend it on the index and the eval set instead.	Beat all three of the above on merit, since it has no advocate

- 2.1** Your case must name four things: the metric your pattern moves, roughly how much you would expect it to move, what it costs on every query, and the single condition under which your own proposal is clearly the wrong choice.
- 2.2** Hear all three. Then rank all four options, including the one nobody argued for. A ranking with no reasoning attached is a vote, not a decision.
- 2.3** Write the refusals. For every option you did not pick, one sentence, in this shape: we refuse X because [metric] is [value], and X moves [that metric], not [the one that is low]. "Not the priority" is a deferral, not a refusal. One of the three can be refused on two independent grounds. Find both.
- 2.4** Name the confirmation. For your chosen action: which number should move, in which direction, and by how much before you would call it a win? Then name the number that must NOT move, and the threshold at which you would roll the change back.
- 2.5** There is one measurement, costing about an afternoon, that would discriminate between two of these three proposals. It has never been run. Name it and say what each possible result would tell you to do.
- 2.6** Pick any one of the patterns you rejected. Change exactly one fact in Sections 5, 6, or 7 so that it becomes the right answer. Name the fact and the new value. This is how you find out whether you rejected it for a reason or by reflex.

12. Round 3: report out

#	State this	What makes it count
1	What you are adding, or that you are adding nothing, and the number that decided it	A number from Sections 3, 5, 6, or 7. Not a principle.
2	What you refuse to add, and the specific reason	Name the metric and its value. "Too complex" is not a reason.
3	Where your group genuinely disagreed, and whether you resolved it	Not optional. In a group of three there is no majority, so a clean consensus means somebody gave in.
4	The measurement you would run before building anything	Precise enough that someone could run it on Monday, and cheap enough that they would.

13. Your deliverable: the Decision Sheet

One per group. Hand it to your facilitator at the end.

Part 1. The call

Option	Add, defer, or refuse	The number that decided it	What it costs per query
Query transformation			
Reranking			
Self-reflective generation			
Add no pattern			

Part 2. The confirmation plan

What we do first	The number that should move, and which way	The number that must not move, and the rollback threshold

14. If your group is stuck

Three levels per question. Take one at a time and stop as soon as you are moving again. None of these contains an answer and reading all three when level one would have done is the only way to waste them.

Question 1.1 | What the combination tells you

Level 1. Look at the stage column in Section 3. Two of these numbers describe what reaches the model and two describe what the model does with it. Which pair is healthy?

Level 2. Section 4's last column, faithfulness row. Read it again. What kind of answer scores well on faithfulness while leaving a user with nothing?

Level 3. If the model is scoring 0.93 on grounding, it is not inventing things. So, whatever the users are unhappy about, the model is not lying to them. That narrows where the problem can be to one stage. Say which and say what the model is most likely doing when recall is poor.

Question 1.2 | Auditing the eval set

Level 1. Take Section 6 one line at a time and ask a single question of each: could this line make a number look better than reality, or worse?

Level 2. Two lines carry most of the weight. One is about who wrote the queries and what she was looking at while she wrote them. The other is about what is entirely absent from the query set.

Level 3. A question written while reading a document inherits that document's vocabulary. Now look at the largest complaint cluster in Section 7 and ask whether those users had the document open. Separately: faithfulness is scored over answers to questions that have answers. If no query in the set is unanswerable, what behavior has the score never had a chance to observe?

Question 1.4 | Predicting the breakdown

Level 1. An average over 40 queries hides whatever populations are inside it. What different kinds of questions are likely in that set?

Level 2. You do not have to invent the categories. Someone else already built a taxonomy of query types, from an entirely different data source, and it is printed in this packet.

Level 3. Section 7's clusters are mostly descriptions of question shapes rather than system properties. Two of them describe a kind of question that a single embedding search would struggle with, and for a structural reason rather than a tuning reason. If one segment scored near where you want and another near the floor, an aggregate of 0.61 is exactly what you would see.

Question 2.2 and 2.3 | Ranking and refusing

Level 1. Put Section 3 next to Section 8. For each proposal, check whether the number it moves is the number that is low.

Level 2. Two of the three proposals target a metric already above 0.8. Say out loud, in plain words, what improving an already healthy metric buys a user.

Level 3. One of the three also makes something in Section 5 worse, and Section 5 already says that thing is missing its target. A refusal that names both the wrong metric and the violated constraint is complete. A refusal that names only one is merely correct.

Question 2.5 | The measurement nobody has run

Level 1. Two of the three proposals both claim to improve what the model finally receives. Under what circumstance would one of them be structurally unable to help at all?

Level 2. Every recall figure you have was measured at one value of k , and Section 6 tells you which. Nothing has ever been measured at any other value.

Level 3. A reranker can only reorder what the candidate search already returned. It cannot promote a chunk that was never a candidate. So, before anyone buys one, what would you need to know about a much wider candidate set, and what would each possible answer tell you to do instead?

15. Vocabulary for this discussion

Term	Working definition in this context
Outlier metric	The one number out of line with the others. It tells you where to look. It does not tell you what you will find, and it does not tell you whether it is trustworthy.
Candidate set	Everything the cheap first-pass search returned, before anything narrows it. A reranker can only reorder this set. It cannot add to it.
Vocabulary mismatch	The user's words and the document's words describe the same thing differently. Embedding search is better at this than keyword search and still not good at it.
Multi-part query	One question containing two or more independent information needs. A single embedding of the whole thing sits between both answers and may be near neither.
Segment	A subpopulation of your eval set. An aggregate can look mediocre while one segment is fine and another is broken, and those two need different fixes.
Guardrail metric	A number you are not trying to improve that you watch to make sure your change did not damage it. A plan without one can trade a problem rather than solve it.
Complexity tax	The cost a pattern imposes on every query forever, not just the cost of building it. Patterns are almost always cheaper to add than to remove.